

Phase 2 - Backend

Time	Start date	Finish date	Status
2 weeks			Not started

Goal: Build the database and the first REST APIs: clubs, events, tags and the approval workflow.

Real-life example: The kitchen is built before the dining hall is opened. Once the kitchen works, any menu (frontend) can use it.

Step checklist

#	Step	Done
1	Write Flyway migration V1: users, roles, profiles, clubs, club_members, events, tags, event_tags	☐
2	Write Flyway migration V2: sample data - 5 clubs, 20 events, tags	☐
3	Create entities with relationships (One-to-One, One-to-Many, Many-to-Many)	☐
4	Create repositories with Spring Data JPA	☐
5	Create DTOs (request / response) - never return entities directly	☐
6	Create services with business rules	☐
7	Event APIs: list with search, filter by tag / club / date, sort and pagination; get by id; create; update	☐
8	Approval workflow: DRAFT -> PENDING_APPROVAL -> PUBLISHED / back to DRAFT with a reason	☐
9	Validation (@Valid) and a global exception handler that returns clean JSON errors	☐
10	Test every API in Postman and check Swagger	☐

Output of this phase

- Database with 8+ tables and sample data
- Working REST APIs documented in Swagger
- Postman collection saved in the repo

Done when (check before moving on)

Check	Passed
GET /api/events?tag=tech&page=0 returns the right events	☐
Submitting and approving an event changes its status correctly	☐
Wrong input returns a 400 error with a clear message	☐

Skills you practise

Spring Boot, REST, JPA relationships, DTO pattern, validation, exception handling, Flyway

Where your work goes

backend\ (entities, repositories, services, controllers, Flyway migrations)

My notes
